

# SAI KUMAR REDDY ANTHIREDDY

AI/ML Engineer · Generative AI · RAG Pipelines · MLOps

Dallas, TX | +1 (972) 710-9106 | [reddysaikumar614@gmail.com](mailto:reddysaikumar614@gmail.com) | [linkedin.com/in/saireddy2110](https://linkedin.com/in/saireddy2110) | [github.com/saianthireddy](https://github.com/saianthireddy)

## PROFESSIONAL SUMMARY

AI/ML Engineer with 3 years of experience building production LLM systems — RAG pipelines, semantic search, and fine-tuned Transformers with LangChain, Hugging Face, PyTorch, and OpenAI APIs. Ships end to end: retrieval architecture and prompt design through containerized deployment on Kubernetes and SageMaker, CI/CD, and model monitoring.

## TECHNICAL SKILLS

**GenAI & LLMs:** LangChain, Hugging Face Transformers, RAG, Prompt Engineering, OpenAI API, LlamaIndex, LangGraph, AI Agents  
**ML, DL & NLP:** PyTorch, TensorFlow, Scikit-learn, Keras, CNNs, RNNs, Transformers, Transfer Learning, BERT, spaCy, Text Classification, Embedding Models, Semantic Search

**MLOps & Vector DBs:** Docker, Kubernetes, MLflow, FastAPI, REST APIs, GitHub Actions, CI/CD, Model Monitoring, Pinecone, FAISS, ChromaDB

**Cloud & Data:** AWS SageMaker, Azure, GCP, Pandas, NumPy, ETL Pipelines, Feature Engineering, SQL, Python, Java

## PROFESSIONAL EXPERIENCE

**AI/ML Engineer** | Teledyne Technologies Inc.

Jul 2025 – Present

- Designed an **Intelligent Technical Knowledge & Support Automation Platform** using RAG over enterprise knowledge sources to automate first-line technical support.
- Built an end-to-end RAG pipeline with **LangChain, OpenAI API, FAISS, and Pinecone** over manuals, SOPs, and support docs; tuned chunking and embedding across Pinecone and ChromaDB to improve retrieval relevance and latency.
- Fine-tuned Hugging Face Transformers (BERT) for intent classification, benchmarked against a keyword-matching baseline.
- Built agentic workflows with **LangGraph and LlamaIndex** to orchestrate retrieval, reasoning, and tool execution.
- Containerized AI services with Docker and deployed inference on **Kubernetes and AWS SageMaker**; automated training and versioning with GitHub Actions, CI/CD, and MLflow for drift-based retraining and monitoring.
- Mentored junior engineers through code reviews; integrated AI features into enterprise apps with product and domain teams.

**Junior Machine Learning Engineer** | Colt Technology Services

May 2021 – Jun 2023

- Built a **Customer Retention Intelligence & Churn Prediction Platform** to flag at-risk accounts for the retention team.
- Cleaned and transformed **250K+ customer records** with Pandas and NumPy; extended ETL pipelines unifying usage, billing, and support data into ML workflows.
- Engineered features and built supervised models with **Scikit-learn and TensorFlow**, evaluated with cross-validation, precision/recall/F1, and confusion-matrix analysis against the incumbent rules-based model.
- Deployed models as **FastAPI REST APIs on AWS**, enabling real-time churn scoring in place of a nightly batch job.

## PORTFOLIO PROJECTS

- Enterprise AI Platform** — [github.com/saianthireddy/enterprise-ai-platform](https://github.com/saianthireddy/enterprise-ai-platform) — Intent router over six task handlers, hybrid search, admin analytics, Next.js, Terraform + Kubernetes.
- RAG Support Automation** — [github.com/saianthireddy/rag-support-automation](https://github.com/saianthireddy/rag-support-automation) — Swappable FAISS/Pinecone stores, citation-grounded generation, FastAPI, and a published retrieval benchmark.
- Enterprise MLOps Platform** — [github.com/saianthireddy/enterprise-mlops-platform](https://github.com/saianthireddy/enterprise-mlops-platform) — Champion/challenger promotion, versioned registry, PSI/KS drift detection, Kubernetes + SageMaker deploy.
- webdocs-mcp** — [github.com/saianthireddy/webdocs-mcp](https://github.com/saianthireddy/webdocs-mcp) — Docs crawler with hybrid BM25 + embedding search over DuckDB, served to LLM agents as an MCP server.
- Customer Churn Prediction** — [github.com/saianthireddy/customer-churn-prediction](https://github.com/saianthireddy/customer-churn-prediction) — LogReg / Gradient Boosting pipelines, cross-validated evaluation, real-time /predict API with risk bands.

## EDUCATION & CERTIFICATION

**M.S. Advanced Data Analytics**, University of North Texas (2025) · **B.S. Computer Science**, Osmania University (2021) · **AWS Certified Solutions Architect – Associate** — [Verify on Credly](https://www.credly.com/badges/ai-solutions-architect-associate)